

A full answer to the Lorentz case in the Orlicz–Lorentz Daugavet problem

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Source problem

The source paper is A. Kaminska, H. J. Lee, and H.-J. Tag, *Daugavet and diameter two properties in Orlicz–Lorentz spaces*, arXiv:2212.12149. After proving that regular weights force the Daugavet Orlicz–Lorentz case to be L_1 , the authors ask:

Let $\gamma = \infty$ and assume that the weight function w is not regular or $\lim_{t \rightarrow 0+} t/W(t) > 0$.
Can the Lorentz space $\Lambda_{1,w}$ have the Daugavet property, DD2P, or DLD2P?

Here $W(t) = \int_0^t w(s) ds$, and

$$\|f\|_{1,w} = \int_0^\infty f^*(t)w(t) dt$$

is the Lorentz norm, with f^* denoting the decreasing rearrangement.

Result

We prove that the only positive case is the trivial L_1 case.

Theorem 1. *Let w be a positive nonincreasing Lorentz weight on $(0, \infty)$, let $W(t) = \int_0^t w$, and assume $W(\infty) = \infty$. On a nonatomic σ -finite measure space,*

$\Lambda_{1,w}$ has the Daugavet property $\iff \Lambda_{1,w}$ has DD2P $\iff \Lambda_{1,w}$ has DLD2P $\iff W(t) = ct$ for all t

Consequently, if w is not a.e. constant, then $\Lambda_{1,w}$ has none of the Daugavet property, DD2P, or DLD2P.

Proof intuition

The previous route one might try is local uniform nonsquareness of normalized indicators. That route is false: tiny high spikes make $\|x \pm u\|$ almost 2.

The right slice is instead the one exposed by the average over the same set. If A has measure a , put $x_A = \chi_A/W(a)$. When w is genuinely nonconstant on $[0, a]$, the functional

$$\Phi_A(f) = \frac{W(a)}{a} \int_A f$$

strongly exposes x_A in the Lorentz norm. The reason is the strict equality case in the weighted Hardy inequality

$$\frac{W(a)}{a} \int_0^a h(t) dt \leq \int_0^a h(t)w(t) dt$$

for decreasing h . Equality forces h to be constant on $[0, a]$ if w is not constant there. A strongly exposed point cannot be a Δ -point, while DLD2P requires every unit vector to be a Δ -point.

The exposing lemma

We use the following elementary equality-stability form of Chebyshev's integral inequality.

Lemma 1 (weighted Hardy exposure). *Let $a > 0$, let w be nonincreasing on $[0, a]$, and put*

$$c = \frac{1}{a} \int_0^a w(t) dt = \frac{W(a)}{a}.$$

Assume that w is not a.e. constant on $[0, a]$. Then there is a constant $\kappa = \kappa(a, w) > 0$ such that every nonnegative nonincreasing integrable function h on $[0, a]$ satisfies

$$\int_0^a h(t)w(t) dt - c \int_0^a h(t) dt \geq \kappa \int_0^a \left| h(t) - \frac{1}{a} \int_0^a h(s) ds \right| w(t) dt.$$

In particular, if h_n is nonnegative and nonincreasing,

$$c \int_0^a h_n \rightarrow 1, \quad \int_0^a h_n(t)w(t) dt \rightarrow 1,$$

then $h_n \rightarrow W(a)^{-1} \chi_{[0, a]}$ in the weighted $L_1(w dt)$ -norm on $[0, a]$.

Proof. The inequality

$$\int_0^a hw \geq \frac{1}{a} \left(\int_0^a h \right) \left(\int_0^a w \right)$$

is Chebyshev's integral inequality, since both h and w are nonincreasing. More precisely,

$$\int_0^a hw - \frac{1}{a} \left(\int_0^a h \right) \left(\int_0^a w \right) = \frac{1}{a} \int_0^a \int_0^t (h(s) - h(t))(w(s) - w(t)) ds dt. \quad (1)$$

The integrand in (1) is nonnegative.

Because w is not a.e. constant, there are two subintervals $I < J \subset [0, a]$ and a number $\eta > 0$ such that $w(s) - w(t) \geq \eta$ for a.e. $s \in I, t \in J$. Hence the left hand side of (1) controls the mean drop of h from I to J . Applying the same argument to a finite chain of intervals joining the parts of $[0, a]$ on which w is above and below its average gives a constant $\kappa > 0$, depending only on a and w , for which the displayed weighted estimate holds. This is the usual quantitative equality case in Chebyshev's inequality for monotone functions; the weighted form follows by using the same finite chain with the measure $w(t) dt$, which is finite on $[0, a]$.

The final assertion follows immediately. The two convergences force $\int_0^a h_n \rightarrow a/W(a)$, and the estimate then forces the distance of h_n from its own average, measured with $w(t) dt$, to tend to 0. $\square$

Lemma 2. *Let A be measurable with $\mu(A) = a$, and suppose w is not a.e. constant on $[0, a]$. Then*

$$x_A = \frac{\chi_A}{W(a)}$$

is strongly exposed in $B_{\Lambda_{1,w}}$ by

$$\Phi_A(f) = \frac{W(a)}{a} \int_A f d\mu.$$

Proof. First $\|\Phi_A\| \leq 1$. Indeed, for $f \in \Lambda_{1,w}$, Hardy–Littlewood rearrangement and Chebyshev’s inequality give

$$|\Phi_A(f)| \leq \frac{W(a)}{a} \int_0^a f^*(t) dt \leq \int_0^a f^*(t) w(t) dt \leq \|f\|_{1,w}.$$

Also $\|x_A\|_{1,w} = 1$ and $\Phi_A(x_A) = 1$.

Let $f_n \in B_{\Lambda_{1,w}}$ and $\Phi_A(f_n) \rightarrow 1$. Put $h_n = f_n^*$. The chain of inequalities above shows that

$$\frac{W(a)}{a} \int_0^a h_n \rightarrow 1, \quad \int_0^a h_n(t) w(t) dt \rightarrow 1, \quad \int_a^\infty h_n(t) w(t) dt \rightarrow 0.$$

By Lemma 1,

$$h_n \longrightarrow W(a)^{-1} \chi_{[0,a]}$$

in the Lorentz norm. Moreover, the first Hardy–Littlewood inequality is also asymptotically an equality:

$$\int_A f_n \rightarrow \int_0^a h_n.$$

Thus the positive mass of f_n is asymptotically carried by A , the negative part on A is asymptotically zero, and no Lorentz norm remains outside A . Consequently

$$\|f_n - x_A\|_{1,w} \rightarrow 0.$$

This is precisely strong exposure of x_A by Φ_A . □

Proof of the theorem

If w is not a.e. constant on $(0, \infty)$, then there is $a > 0$ such that w is not a.e. constant on $[0, a]$. Choose a measurable set A of measure a . By Lemma 2, the unit vector $x_A = \chi_A/W(a)$ is strongly exposed by a norm-one functional.

A strongly exposed point cannot be a Δ -point: for sufficiently small $\alpha > 0$, the slice

$$S(B_X, \Phi_A, \alpha) = \{y \in B_X : \Phi_A(y) > 1 - \alpha\}$$

has arbitrarily small norm diameter around x_A , and therefore cannot contain points y with $\|x_A - y\| > 2 - \varepsilon$. Hence x_A is not a Δ -point.

The diametral local diameter two property is equivalent to every point of the unit sphere being a Δ -point. Therefore $\Lambda_{1,w}$ does not have DLD2P. Since

$$\text{Daugavet property} \Rightarrow \text{DD2P} \Rightarrow \text{DLD2P},$$

it has neither DD2P nor the Daugavet property.

Conversely, if w is a.e. constant, say $w = c > 0$, then

$$\|f\|_{1,w} = c \int_0^\infty f^*(t) dt = c \|f\|_{L_1}.$$

Thus $\Lambda_{1,w}$ is isometric, up to the scalar c , to the usual nonatomic L_1 -space. Nonatomic L_1 has the Daugavet property, hence also DD2P and DLD2P. This proves Theorem 1.

Novelty and verification notes

The run indexes contained one earlier attempt for this source problem, `2212.12149_lorentz_daugavet_nonregular`. That attempt correctly observed that the local-uniform-nonsquareness route fails because of tiny high spikes. The present proof uses a different slice and a strong exposure argument.

Bounded phrase searches for the exact Lorentz problem in the earlier attempt found the source paper itself and nearby rearrangement-invariant Daugavet literature, but no later exact answer. The main point for human verification is Lemma 1, the quantitative equality case in Chebyshev's inequality for monotone functions.

References

- [1] A. Kaminska, H. J. Lee, and H.-J. Tag, *Daugavet and diameter two properties in Orlicz–Lorentz spaces*, arXiv:2212.12149.
- [2] V. Kadets, R. Shvidkoy, G. Sirotkin, and D. Werner, *Banach spaces with the Daugavet property*, Trans. Amer. Math. Soc. 352 (2000), 855–873.